A Rank-Sum Stock Screener on S&P 500 Fundamentals

stock-ranker

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Abstract

A Java program pulls quarterly financial statements, company overviews, news sentiment and earnings surprises from the Alpha Vantage API for a list of S&P 500 tickers, ranks every company on 50 metrics, and combines the ranks into one weighted score. The 50 highest-scoring companies returned +17.4% on average (median +11.4%) between the entry and exit dates of the test window; 35 of 50 rose and 15 fell.

1 Method

For every ticker the program collects:

- **Income statement, balance sheet, cash flow** – 29 line items, each averaged over the last 12 quarters;
- **Company overview** – 15 valuation, profitability, risk and analyst-rating fields (P/E, P/B, P/S, dividend yield, ROE, beta, ...);
- **Derived ratios** – gross and net margin, current ratio, debt-to-equity;
- **News sentiment** – mean score of up to 25 articles from the last six months;
- **Earnings surprise** – mean EPS surprise over the last 8 quarters.

For each metric m with weight w_m, the N tickers that report it are sorted best-first (descending, or ascending for "lower is better" metrics such as P/E, beta, liabilities or analyst sell ratings). The company in position i (starting at 0) receives $w_m(N - i)$ points, and a company's score is the sum over all metrics:

$$S_c = \sum_m w_m(N - \mathrm{rank}_m(c)).$$

Weights range from 1 to 6. News sentiment and earnings surprise carry the largest weight (6); net income, R&D, earnings growth, beta, strong-buy ratings, investing cash flow and long-term debt issuance carry 3.

2 Results

Table 1 lists the 50 companies in the order the screener ranked them, with the entry and exit prices and the simple return. Figure 1 shows the same returns sorted from best to worst.

Mean return (equal-weight portfolio)	+17.4%
Median return	+11.4%
Standard deviation	30.7%
Winners / losers	35 / 15 (70% hit rate)
Best pick	INTC (+92.9%)
Worst pick	UNH (-42.6%)
Mean of ranks 1–10 vs. 11–50	+19.6% vs. +16.9%

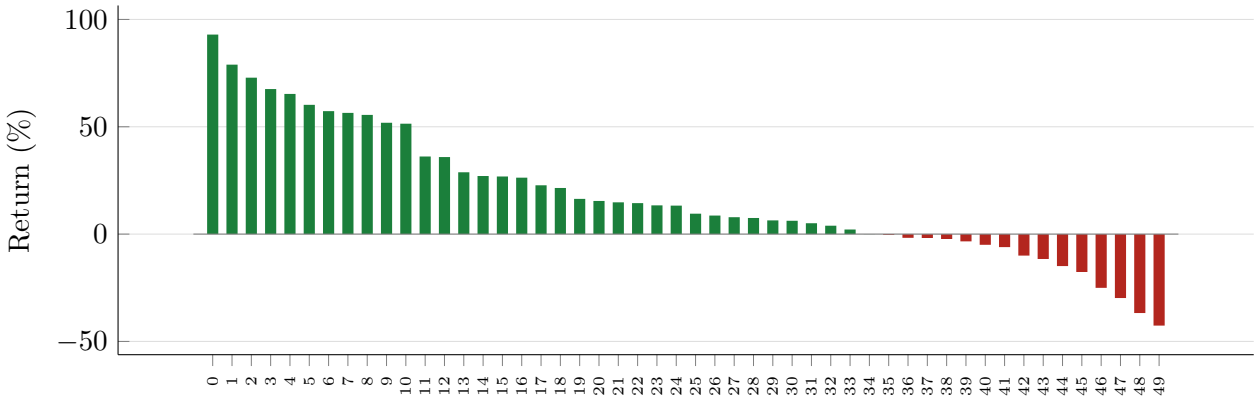


Figure 1: Return of each pick, sorted.

Table 1: The 50 top-ranked companies. Return in %.

#	Ticker	Entry	Exit	Ret.	#	Ticker	Entry	Exit	Ret.
1	F	9.71	13.22	+36.1	26	DELL	91.46	153.26	+67.6
2	MRK	112.31	84.20	-25.0	27	CSCO	45.83	72.08	+57.3
3	XOM	114.77	114.83	+0.1	28	APD	279.21	237.63	-14.9
4	TMO	598.03	568.27	-5.0	29	CMCSA	39.35	27.63	-29.8
5	AMZN	161.02	250.44	+55.5	30	ABT	109.76	124.33	+13.3
6	JNJ	161.25	186.15	+15.4	31	BMY	46.85	46.04	-1.7
7	JPM	194.90	312.27	+60.2	32	AIG	72.14	77.82	+7.9
8	WMT	67.59	102.65	+51.9	33	HD	348.64	374.83	+7.5
9	TXN	181.19	163.06	-10.0	34	INTC	19.71	38.03	+92.9
10	NOC	492.63	573.48	+16.4	35	MDT	82.04	89.84	+9.5
11	LLY	775.26	951.52	+22.7	36	PEP	172.62	142.16	-17.6
12	SO	86.57	92.10	+6.4	37	CB	273.41	284.09	+3.9
13	VZ	39.80	39.66	-0.4	38	PCAR	95.17	99.99	+5.1
14	AMGN	325.22	319.32	-1.8	39	CRM	262.91	254.03	-3.4
15	QCOM	157.95	179.10	+13.4	40	ALL	191.60	195.70	+2.1
16	GS	459.02	793.51	+72.9	41	T	26.24	24.65	-6.1
17	UNH	569.96	327.03	-42.6	42	ACN	390.22	246.73	-36.8
18	CAT	316.80	566.89	+78.9	43	BAC	46.01	52.82	+14.8
19	IBM	183.31	303.01	+65.3	44	CVX	157.23	153.65	-2.3
20	MSFT	395.15	508.94	+28.8	45	C	83.94	101.98	+21.5
21	T	19.40	24.65	+27.1	46	GOOG	187.13	283.37	+51.4
22	AXP	233.54	365.41	+56.5	47	PG	166.49	147.19	-11.6
23	BLK	854.83	1079.56	+26.3	48	SCHW	82.01	93.84	+14.4
24	DE	349.14	474.44	+35.9	49	EIX	52.42	56.95	+8.6
25	AAPL	213.31	270.56	+26.8	50	COF	209.82	222.85	+6.2

AT&T (T) appears twice (ranks 21 and 41) with different entry prices, as in the original output; both rows are kept in the statistics.

3 Limitations

- One period, no benchmark: the result should be compared with the S&P 500 over the same window before drawing conclusions.
- Look-ahead bias: Alpha Vantage returns *current* fundamentals, so a ranking run after the entry date partly uses information that was not available at entry.
- The ticker list is an older S&P 500 snapshot; companies that were delisted or renamed simply receive no data (survivorship bias).
- Size-type metrics (revenue, total assets, market cap) favour large companies, so the screen

leans towards mega-caps.

This is a student project, not investment advice.